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. Student Data Analysis and Operations

57:30

e a Python program that takes the file name of a CSV file containing student ils, including roll numbers and their marks in three subjects as input, reads the , and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject

Explorer

Operatio...

Submi

```python
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details
```

Average time **0.019 s** 19.00 ms

Maximum time **0.019 s** 19.00 ms

1 out of 1 shown test case(s) passed

Test case 1 19 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]

Terminal Test cases

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Numpy Array Operations

10:18

e a Python program to create a NumPy array based on user input and display following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

ume all input elements are valid integers.

ut Format:

- The first line contains two space-separated integers, r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

put Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

numpyarr...

```python
import numpy as np
r,c = map(int,input().split())

data=[]
if r==0 and c==0:
```

Average time **0.035 s** 34.75 ms

Maximum time **0.056 s** 56.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 48 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]

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Numpy: Matrix Operations

07:39

given code takes two 3×3 matrices, matrix_a, and matrix_b, as input the user and converts them into NumPy arrays.

k:

are required to compute and display the results of the following matrix rations:

1. **Addition** ($\text{matrix_a} + \text{matrix_b}$)
2. **Subtraction** ($\text{matrix_a} - \text{matrix_b}$)
3. **Element-wise Multiplication** ($\text{matrix_a} * \text{matrix_b}$)
4. **Matrix Multiplication** ($\text{matrix_a} \cdot \text{matrix_b}$)
5. **Transpose of Matrix A**

ut Format:

- The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

put Format:

Explorer

matrixOp...

Submi

```python
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split()))
```

Average time	Maximum time
0.071 s	**0.074 s**
71.00 ms	74.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 72 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1

Terminal Test cases

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Numpy: Horizontal and Vertical Stacking of Arrays

07:18

are given two arrays `arr1` and `arr2`. You need to perform horizontal and cal stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

ut Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

put Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases +

Explorer

stacking.py

Submi

```python
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i
```

Average time	Maximum time
0.070 s	**0.083 s**
70.25 ms	83.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 66 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6

Terminal Test cases

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Numpy: Custom Sequence Generation

02:05

e a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

program should then generate a sequence using `numpy` based on these inputs print the generated sequence.

ut Format:

- The user will input three integer values: start, stop, and step, each on a new line.

put Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases +

customS...

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
```

Average time	Maximum time
0.035 s	0.037 s
35.00 ms	37.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 37 ms Debug

Expected output	Actual output
3	3
10	10
2	2
[3 5 7 9]	[3 5 7 9]

Test case 2 35 ms

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. Numpy: Arithmetic and Statistical Operations, Mathematic... 04:51

are given two arrays A and B. Your task is to complete the function ay_operations, which will convert these lists into NumPy arrays and orm the following operations:

rithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

tatistical Operations:

- Calculate the mean, median, and standard deviation of array A.

itwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

ut Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

put Format:

Explorer

different...

Submi

```
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
```

Average time 0.032 s 31.67 ms

Maximum time 0.036 s 36.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 36 ms Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5

Terminal Test cases

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. Numpy: Copying and Viewing Arrays

02:13

given code takes a list of integers as input and converts it into a NumPy array.
r task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

r completing these steps, observe how modifying the view affects the
ginal_array, while modifying the copy does not.

ut Format:

- A single line of space-separated integers.

put Format:

- After modifying the view:

```
iginal array after modifying view: <original_array>
ew array: <view_array>
```

- After modifying the copy:

```
iginal array after modifying copy: <original_array>
```

Explorer

copyAnd...

Submi

```
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
```

Average time	Maximum time
0.014 s	**0.016 s**
14.25 ms	16.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 16 ms Debug

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]

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. Numpy: Searching, Sorting, Counting, Broadcasting 10:31

given code in the editor takes a single array, `array1`, as space-separated gers as input from the user.

itionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

need to complete the code to perform the following operations:

earching: Find the indices where `search_value` appears in `array1` and t these indices.

ounting: Count how many times `count_value` appears in `array1` and print count.

roadcasting: Add `broadcast_value` to each element of `array1` using adcasting, and print the resulting array.

orting: Sort `array1` in ascending order and print the sorted array.

ut Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.

Explorer

arrayOpe...

Submi

```python
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

```

Average time **0.046 s** 46.00 ms

Maximum time **0.050 s** 50.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 50 ms Debug

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3

Terminal Test cases

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